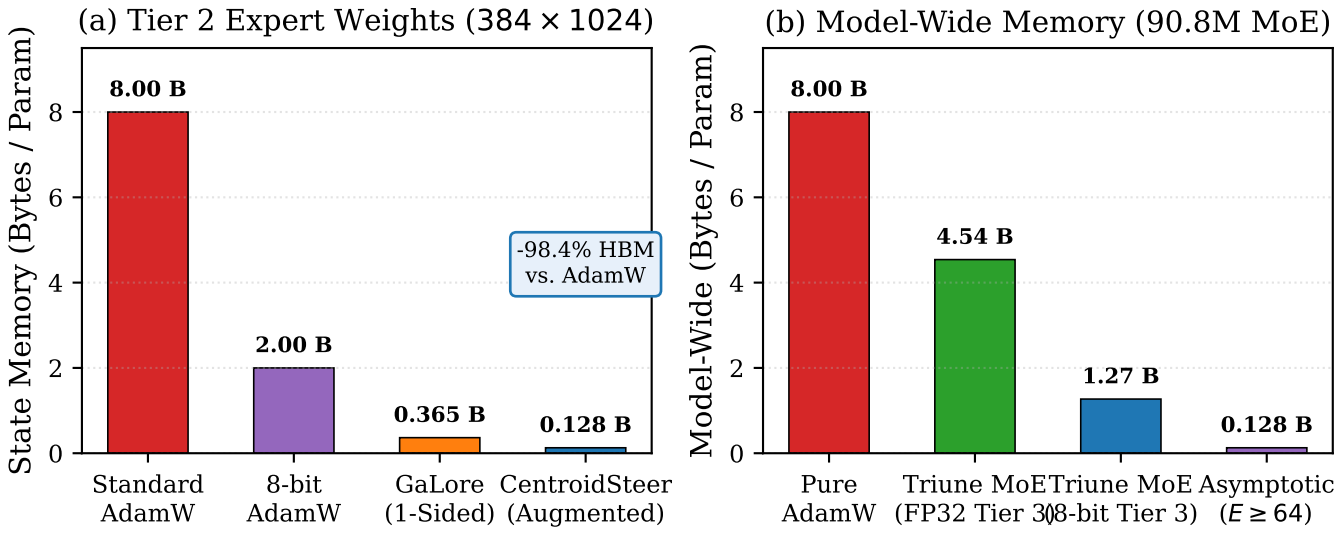
(a) Tier 2 Expert Weights (384 × 1024)
State Memory (Bytes / Param)
8.00 B
2.00 B
0.365 B
0.128 B
-98.4% HBM
vs. AdamW
Standard AdamW
8-bit AdamW
GaLore (1-Sided)
CentroidSteer (Augmented)
(b) Model-Wide Memory (90.8M MoE)
Model-Wide (Bytes / Param)
8.00 B
4.54 B
1.27 B
0.128 B
Pure AdamW
Triune MoE (FP32 Tier 3)
Triune MoE (8-bit Tier 3)
Asymptotic (E ≥ 64)